Product Specification

AU OPTRONICS CORPORATION

( ) Preliminary Specifications
(✓) Final Specifications

Module	14.0" QHD+ 16:10 Color TFT-LCD with LED Backlight design
Model Name	B140QAN03.H
H/W	1A
Note ( )	*LED Backlight with driving circuit design*

Customer	Date	Approved by	Date
		Bryant Chiu	02/15/2023
Checked & Approved by	Date	Prepared by	Date
		Katrina Chen	02/15/2023

Note: This Specification is subject to change without notice.

AU Optronics corporation

Contents

Record of Revision

Version and Date	Page	Old description	New Description	Remark
0.1 2022/08/24	All	First Edition for Customer		
1.0 2023/02/15	All	Final spec.		
1.1 2023/07/07	33	Revise 10.2 Notes		

1. Handling Precautions

1) Since front polarizer is easily damaged, pay attention not to scratch it.
2) Be sure to turn off power supply when inserting or disconnecting from input connector.
3) Wipe off water drop immediately. Long contact with water may cause discoloration or spots.
4) When the panel surface is soiled, wipe it with absorbent cotton or other soft cloth.
5) Since the panel is made of glass, it may break or crack if dropped or bumped on hard surface.
6) Since CMOS LSI is used in this module, take care of static electricity and insure human earth when handling.
7) Do not open nor modify the Module Assembly.
8) Do not press the reflector sheet at the back of the module to any directions.
9) At the insertion or removal of the Signal Interface Connector, be sure not to rotate nor tilt the Interface Connector of the TFT Module.
11) After installation of the TFT Module into an enclosure (Notebook PC Bezel, for example), do not twist nor bend the TFT Module even momentary. At designing the enclosure, it should be taken into consideration that no bending/twisting forces are applied to the TFT Module from outside. Otherwise the TFT Module may be damaged.
12) Small amount of materials having no flammability grade is used in the LCD module. The LCD module should be supplied by power complied with requirements of Limited Power Source (IEC60950 or UL1950), or be applied exemption.
13) Disconnecting power supply before handling LCD modules, it can prevent electric shock, DO NOT TOUCH the electrode parts, cables, connectors and LED circuit part of TFT module that a LED light bar build in as a light source of back light unit. It can prevent electrostatic breakdown.

2. General Description

B140QAN03.H is a Color Active Matrix Liquid Crystal Display composed of a TFT LCD panel, a driver circuit, and LED backlight system. The screen format is intended to support the 16:10 QHD+, 2560(H) x1600(V) screen and 16.7M colors (true RGB 8-bits data driver) with LED backlight driving circuit. All input signals are eDP(Embedded DisplayPort) interface compatible.

B140QAN03.H is designed for a display unit of notebook style personal computer and industrial machine.

2.1 General Specification

The following items are characteristics summary on the table at 25 ℃ condition:

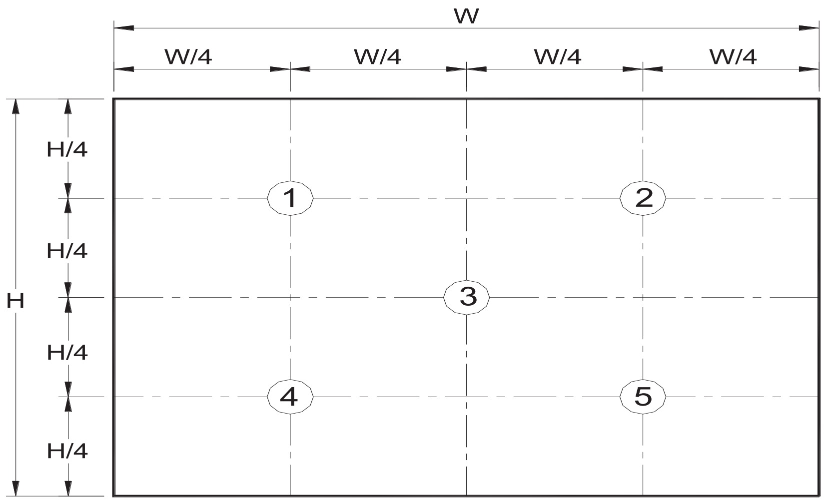
Items	Unit	Specifications			
Screen Diagonal	[mm]	355.654			
Active Area	[mm]	301.594 X188.496			
Pixels H x V		2560x3(RGB) x 1600			
Pixel Pitch	[mm]	0.1178X0.1178			
Pixel Format		R.G.B. Vertical Stripe			
Display Mode		Normally Black (AHVA)			
White Luminance (ILED=19.5mA) **(Note: ILED is LED current)**	[cd/m²]	500 typ. (5 points average) 425 min. (5 points average)			
Luminance Uniformity		1.25 max. (5 points)			
Contrast Ratio		1200:1 typ.			
Response Time	[ms]	9 typ / 13 Max			
Nominal Input Voltage VDD	[Volt]	+3.3 typ.			
Power Consumption	[Watt]	6.15 max@mosaic.(Include Logic& Blu power)			
Weight	[Grams]	210 max.			
Physical Size **Include bracket**	[mm]		Min.	Typ.	Max.
		Length	306.994	307.294	307.594
		Width	198.946	199.246	199.546
		Thickness	-	-	2.0 (Panel) 3.8 (PCBA)
Electrical Interface		4 Lane eDP 1.4 (5.4G)			
Glass Thickness	[mm]	0.2			
Surface Treatment		Anti Glare, Hardness 3H,			
Support Color		16.7M colors (RGB 8-bit)			
Temperature Range Operating Storage (Non-Operating)	 [°C] [°C]	 0 to +50 -20 to +60			
RoHS Compliance		RoHS Compliance			

2.2 Optical Characteristics

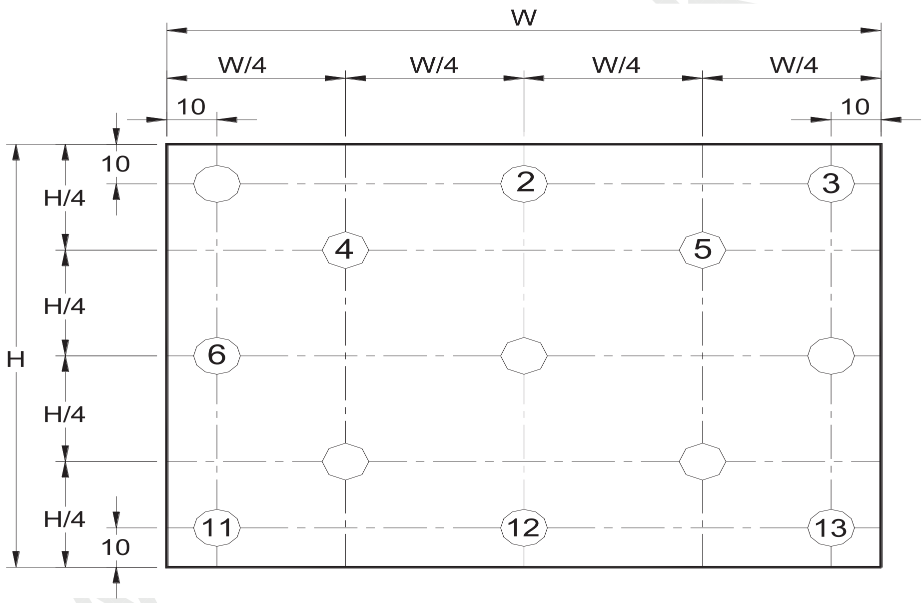
The optical characteristics are measured under stable conditions at 25℃ (Room Temperature) :

Item	Symbol	Conditions	Min.	Typ.	Max.	Unit	Note
White Luminance I_{LED}=19.5mA		5 points average	425	500	-	cd/m²	1, 4, 5.
Viewing Angle	θ_R	Horizontal (Right) CR = 10	80	89	-	degree	4, 9
	θ_L	Horizontal (Left) CR = 10	80	89	-		
	ψ_H	Vertical (Upper) CR = 10	80	89	-		
	ψ_L	Vertical (Lower) CR = 10	80	89	-		
Luminance Uniformity	δ_{5P}	5 Points	-	-	1.25		1, 3, 4
Luminance Uniformity	δ_{13P}	13 Points	-	-	1.6		2, 3, 4
Contrast Ratio	CR		1000	1200	-		4, 6
Cross talk	%				4		4, 7
Response Time	T_{RT}	Rising + Falling	-	9	13	msec	4, 8
Color / Chromaticity Coordinates — Red	Rx	CIE 1931	0.649	0.679	0.709		4
Red	Ry		0.288	0.318	0.348		
Green	Gx		0.236	0.266	0.296		
Green	Gy		0.665	0.695	0.725		
Blue	Bx		0.122	0.152	0.182		
Blue	By		0.02	0.050	0.08		
White	Wx		0.283	0.313	0.343		
White	Wy		0.299	0.329	0.359		
DCI-P3	%		-	100	-		

Note 1: 5 points position (Ref: Active area)



Note 2: 13 points position (Ref: Active area)



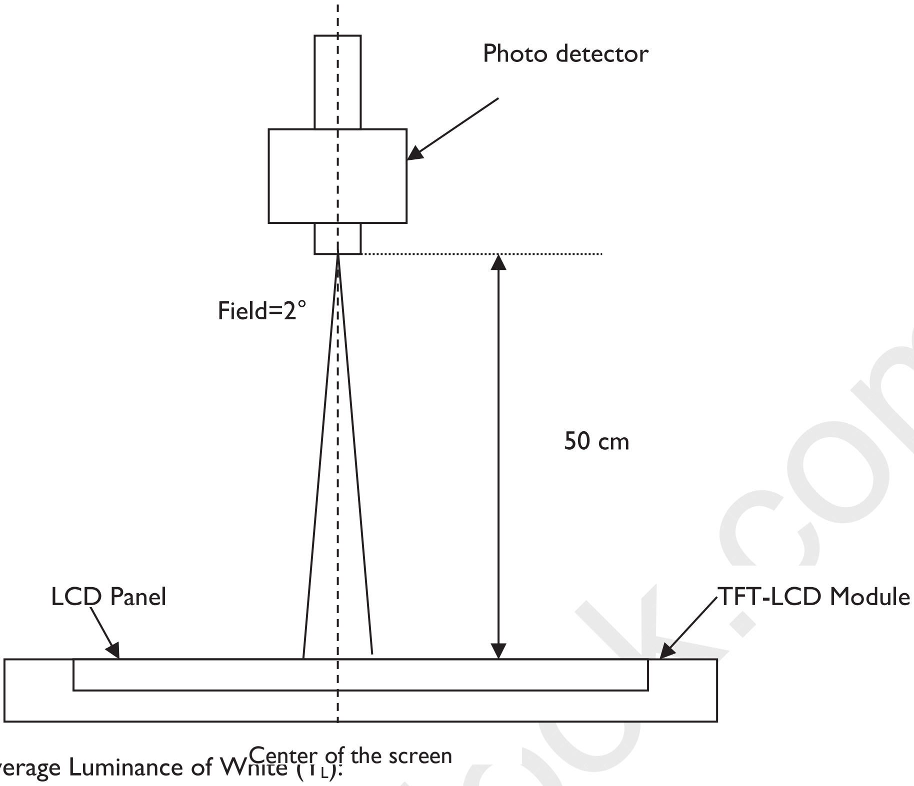
Note 3: The luminance uniformity of 5 or13 points is defined by dividing the maximum luminance values by the minimum test point luminance

$$\delta_{W5} = \frac{\text{Maximum Brightness of five points}}{\text{Minimum Brightness of five points}}$$

$$\delta_{W13} = \frac{\text{Maximum Brightness of thirteen points}}{\text{Minimum Brightness of thirteen points}}$$

Note 4: Measurement method

The LCD module should be stabilized at given temperature for 30 minutes to avoid abrupt temperature change during measuring. In order to stabilize the luminance, the measurement should be executed after lighting Backlight for 30 minutes in a stable, windless and dark room, and it should be measured in the center of screen.



Note 5： Definition of Average Luminance of White (Y_L):

Measure the luminance of gray level 63 at 5 points，Y_L = [L (1)+ L (2)+ L (3)+ L (4)+ L (5)] / 5

L (x) is corresponding to the luminance of the point X at Figure in Note (1).

Note 6： Definition of contrast ratio:

Contrast ratio is calculated with the following formula.

$$\text{Contrast ratio (CR)} = \frac{\text{Brightness on the "White" state}}{\text{Brightness on the "Black" state}}$$

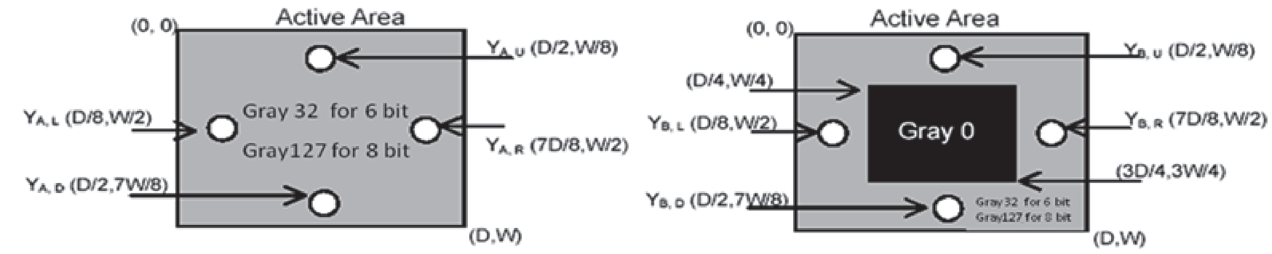
Note 7： Definition of Cross Talk (CT)

CT = | Y_B – Y_A | / Y_A × 100 (%)

Where

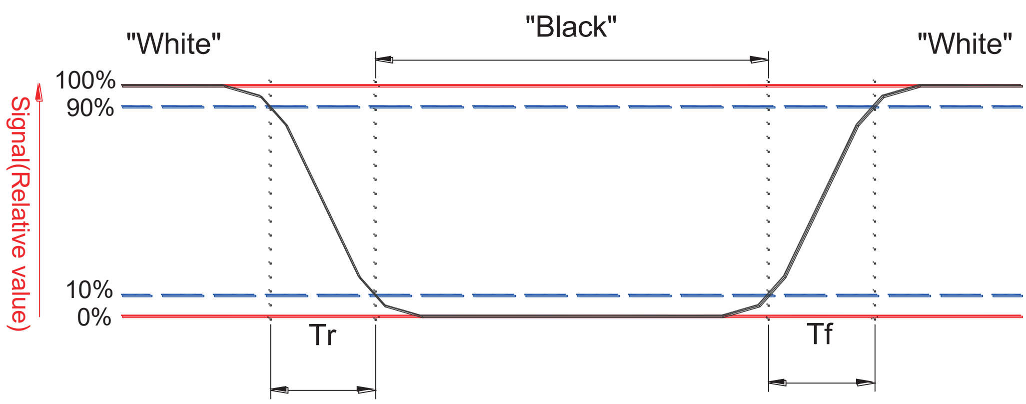
Y_A = Luminance of measured location without gray level 0 pattern (cd/m2)

Y_B = Luminance of measured location with gray level 0 pattern (cd/m2)



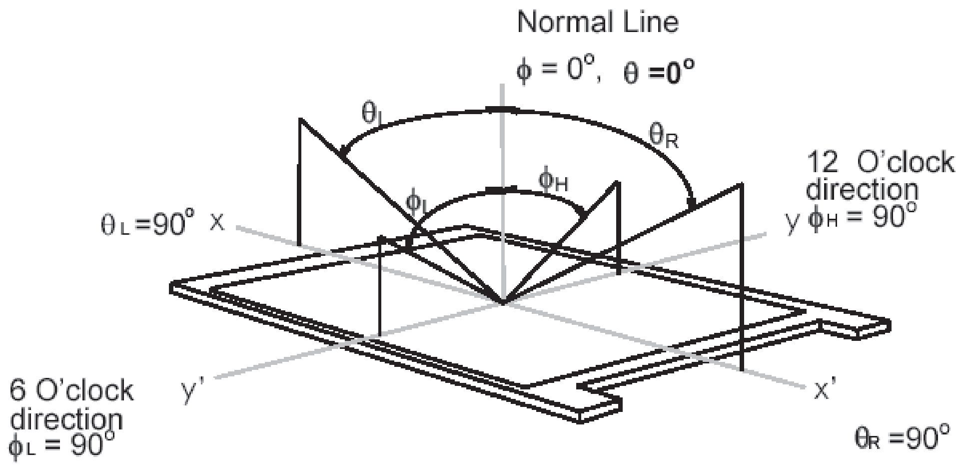
Note 8: Definition of response time:

The output signals of BM-7 or equivalent are measured when the input signals are changed from "Black" to "White" (falling time) and from "White" to "Black" (rising time), respectively. The response time interval between the 10% and 90% of amplitudes. Refer to figure as below.



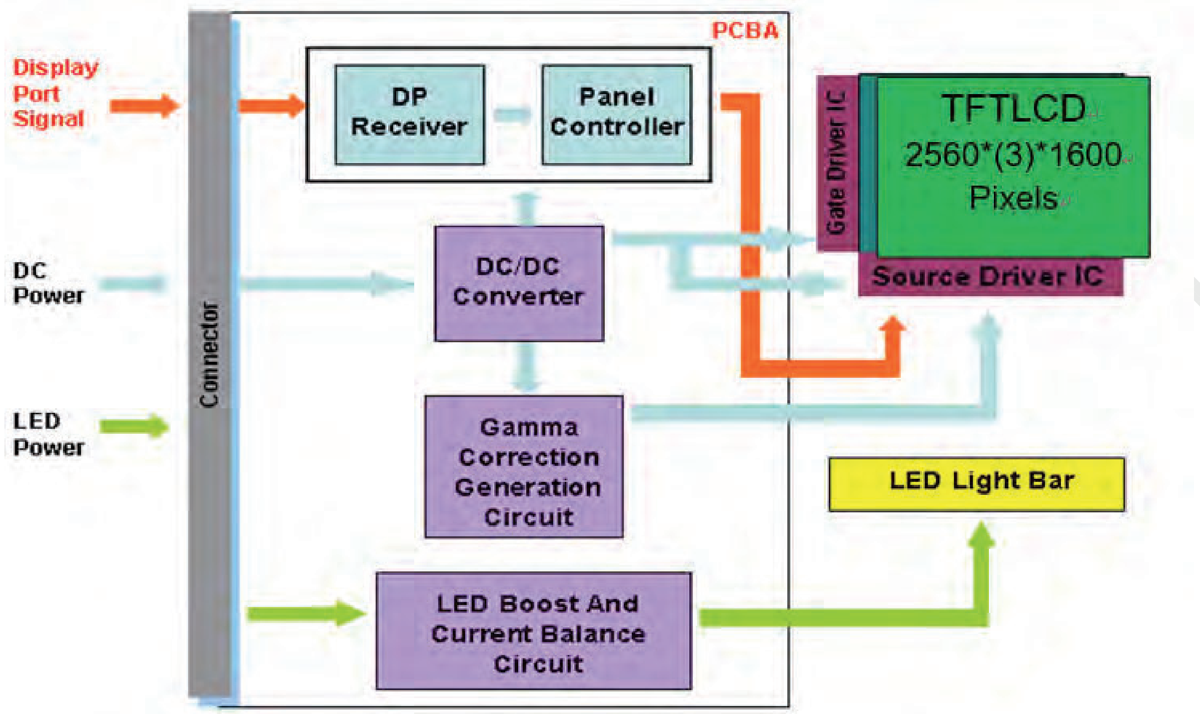
Note 9. Definition of viewing angle

Viewing angle is the measurement of contrast ratio ≧10, at the screen center, over a 180° horizontal and 180° vertical range (off-normal viewing angles). The 180° viewing angle range is broken down as follows; 90° (θ) horizontal left and right and 90° (Φ) vertical, high (up) and low (down). The measurement direction is typically perpendicular to the display surface with the screen rotated about its center to develop the desired measurement viewing angle.



3. Functional Block Diagram

The following diagram shows the functional block of the 14.0 inches wide Color TFT/LCD 40 Pin



4. Absolute Maximum Ratings

An absolute maximum rating of the module is as following:

4.1 Absolute Ratings of TFT LCD Module

Item	Symbol	Min	Max	Unit	Conditions
Logic/LCD Drive Voltage	Vin	-0.3	+4.0	[Volt]	Note 1,2

4.2 Absolute Ratings of Environment

Item	Symbol	Min	Max	Unit	Conditions
Operating Temperature	TOP	0	+50	[°C]	Note 4
Operating Humidity	HOP	5	95	[%RH]	Note 4
Storage Temperature	TST	-20	+60	[°C]	Note 4
Storage Humidity	HST	5	95	[%RH]	Note 4

Note 1: At Ta (25℃)

Note 2: Permanent damage to the device may occur if exceed maximum values

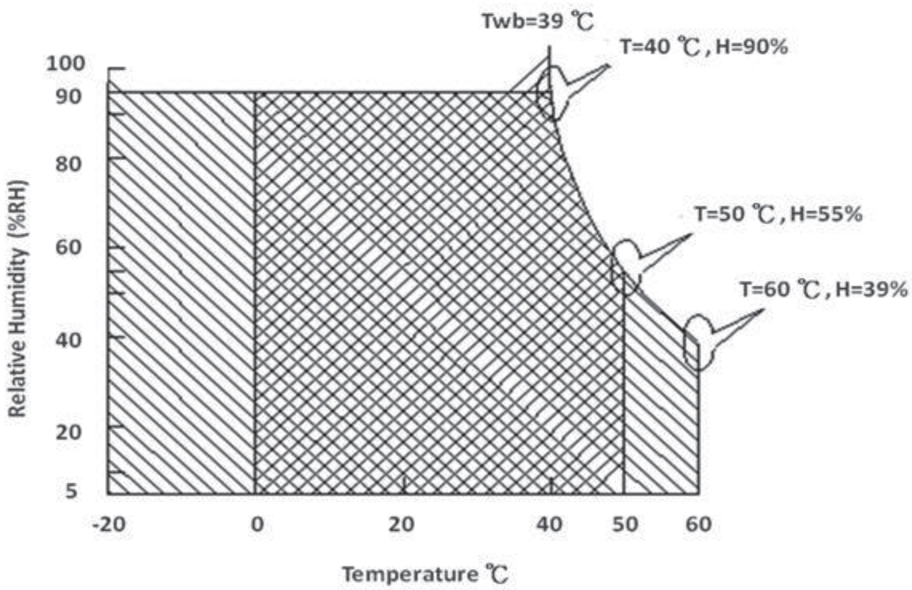
Note 3: LED specification refer to section 5.2

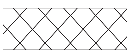
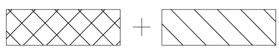
Note 4: For quality performance, please refer to AUO IIS (Incoming Inspection Standard).

Note 5: The packing material of system forbid to involve ammonium component

Note 6: The reliability test conditions of system do not exceed the verified conditions of TFT module

Note 7: Be sure the panel test condition do not exceed the component limitation of TFT module(TN Liquid crystal , for example)



Operating Range  Storage Range 

5. Electrical Characteristics

5.1 TFT LCD Module

5.1.1 Power Specification

Input power specifications are as follows;

The power specification are measured under 25℃ and frame frequency under 60Hz

Symble	Parameter	Min	Typ	Max	Units	Note
VDD	Logic/LCD Drive Voltage	3.0	3.3	3.6	[Volt]	
PDD	VDD Power	-	0.95	1.6	[Watt]	Note 1
PDD(worst)	VDD Power	-	-	2.7	[Watt]	Note 1
IDD	IDD Current(RMS)	-	-	533.3	[mA]	Note 1
IDD(worst pattern)	IDD Current(RMS)	-	-	900	[mA]	Note 1
IRush	Inrush Current	-	-	2000	[mA]	Note 2
Ipeak	Peak Current	-	-	NA	[mA]	Note 2
VDDrp	Allowable Logic/LCD Driver Ripple Voltage	-	-	100	[mV] p-p	

Note 1 : PDD(typ)@ mosaic pattern Maximum Power; PDD(Max)@ R/G/B pattern Maximum Power

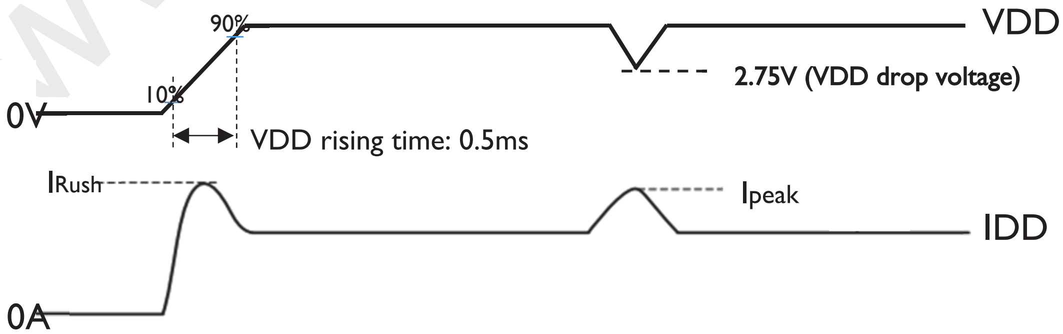
IDD(Max)=PDD(Max) / VDD(Min)

Worst pattern: H-stripe pattern

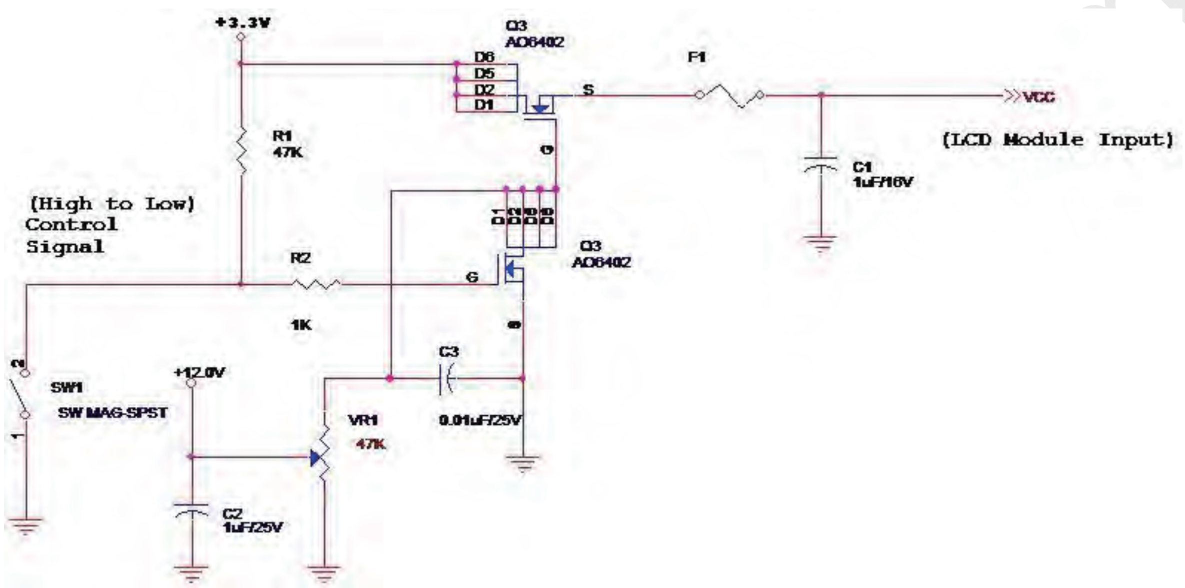
Note 2 :

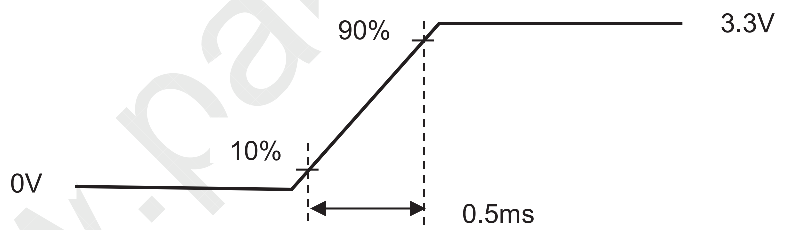
a. IRush: VDD measured rising timing @ 0.5ms

b. IPeak: VDD drop Voltage ≥ 2.75V at this Ipeak current



IRush Measure Condition:





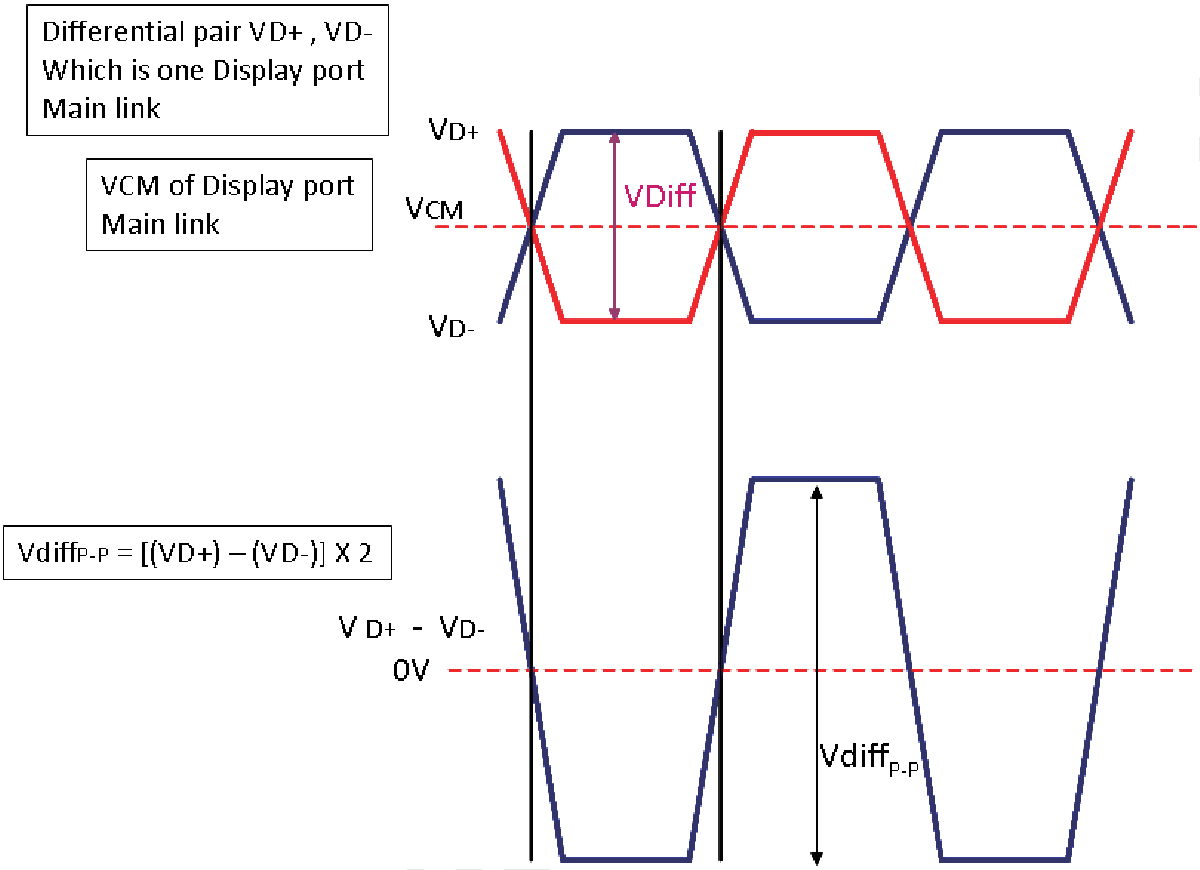
Vin rising time

5.1.2 Signal Electrical Characteristics

Input signals shall be low or High-impedance state when VDD is off.

Signal electrical characteristics are as follows;

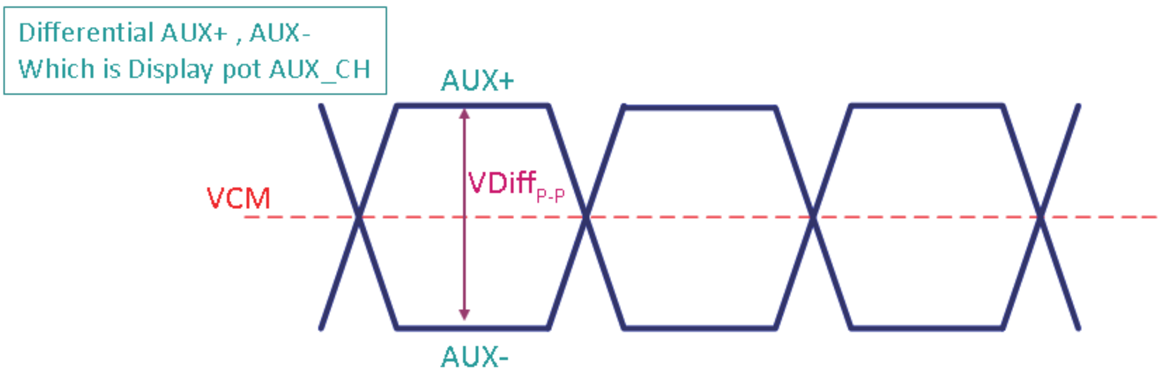
Display Port main link signal:



Display port main link					
		Min	Typ	Max	unit
VCM	RX input DC Common Mode Voltage	0		2.0	V
$VDiff_{P-P}$	Peak-to-peak Voltage at a receiving Device	HBR:70		1320	mV

Follow as VESA display port standard V1.4

Display Port AUX_CH signal:



Display port AUX_CH					
		Min	Typ	Max	unit
VCM	AUX DC Common Mode Voltage	0		1.2	V
$VDiff_{P-P}$	AUX Peak-to-peak Voltage at a receiving Device	140		1360	mV

Fallow as VESA display port standard V1.3

Display Port VHPD signal:

Display port VHPD					
		Min	Typ	Max	unit
VHPD	HPD Voltage	2.25		3.6	V

Fallow as VESA display port standard V1.3

5.2 Backlight Unit

5.2.1 LED characteristics

Parameter	Symbol	Min	Typ	Max	Units	Condition
Backlight Power Consumption	PLED	-	-	5.2	[Watt]	(Ta=25℃), Note 1 Vin =12V
LED Life-Time	N/A	15,000	-	-	Hour	(Ta=25℃), Note 2 I_F=19.5 mA

Note 1: Calculator value for reference P_{LED} = VF (Normal Distribution) * IF (Normal Distribution) / Efficiency

Note 2: The LED life-time define as the estimated time to 50% degradation of initial luminous.

5.2.2 Backlight input signal characteristics

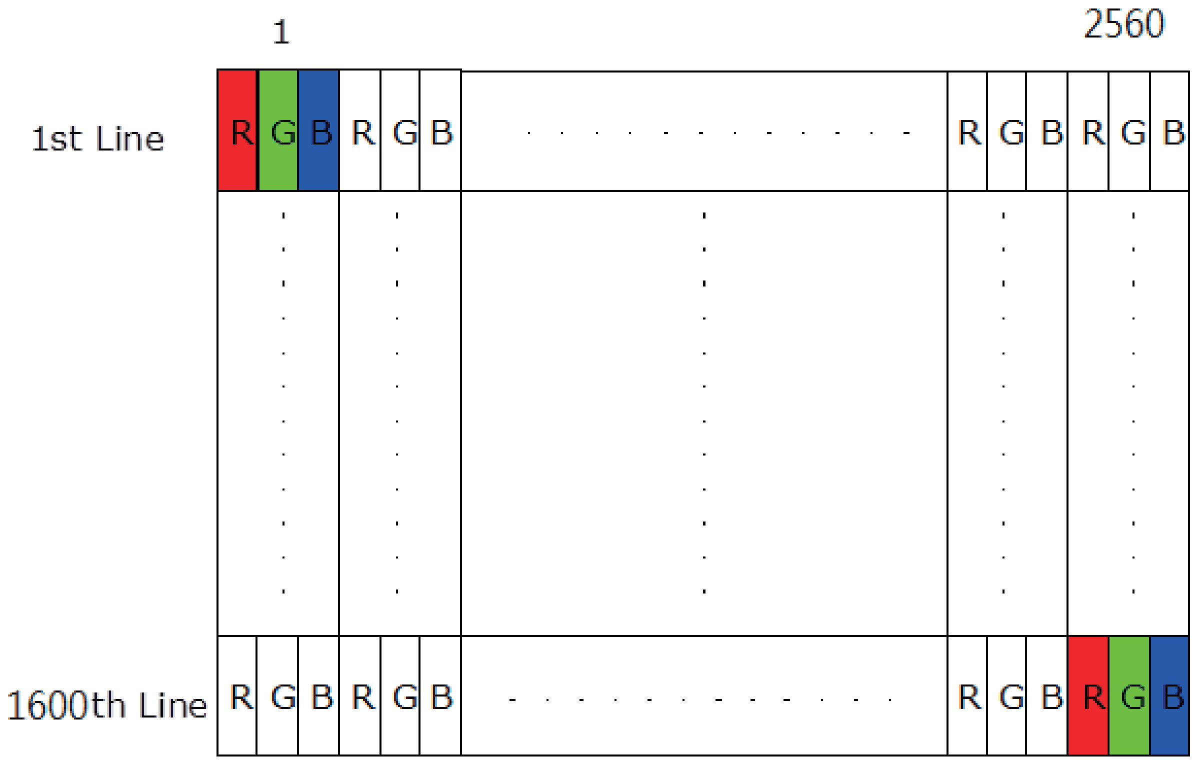
Parameter	Symbol	Min	Typ	Max	Units	Remark
LED Power Supply	VLED	5.0	12.0	21.0	[Volt]	Define as Connector Interface (Ta=25℃)
LED Enable Input High Level	VLED_EN	2.5	-	5.5	[Volt]	
LED Enable Input Low Level		-	-	0.5	[Volt]	
PWM Logic Input High Level	VPWM_EN	2.5	-	5.5	[Volt]	
PWM Logic Input Low Level		-	-	0.5	[Volt]	
PWM Input Frequency	FPWM	200	1K	10K	Hz	
PWM Duty Ratio	Duty	5	--	100	%	

Note 1 : Recommend system pull up/down resistor no bigger than 10kohm

6. Signal Interface Characteristic

6.1 Pixel Format Image

Following figure shows the relationship of the input signals and LCD pixel format.



6.2 Integration Interface Requirement

6.2.1 Connector Description

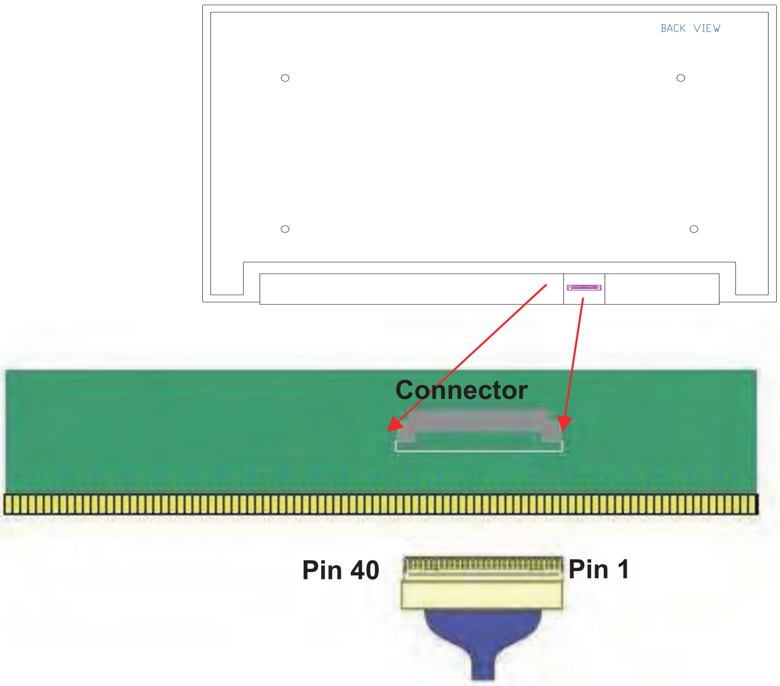
Physical interface is described as for the connector on module.
These connectors are capable of accommodating the following signals and will be following components.

Connector Name / Designation	For Signal Connector
Manufacturer	IPEX or compatible
Type / Part Number	20765-040E-11A or compatible
Mating Housing/Part Number	20704-040T-13 or compatible

6.2.2 Pin Assignment (4 Lane)

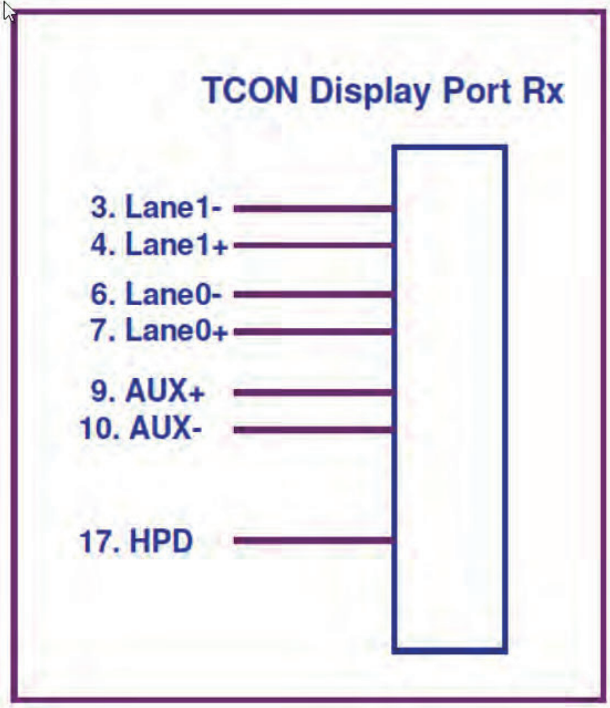
eDP lane is a differential signal technology for LCD interface and high speed data transfer device.

PIN NO	Symbol	Function
1	DDS_SCL	DDS I2C SCL(Note1) _parallel 4.7kohm pull 1.8V
2	H_GND	High Speed Ground
3	Lane3_N	Comp Signal Lane3
4	Lane3_P	True Signal Link Lane 3
5	H_GND	High Speed Ground
6	Lane2_N	Comp Signal Link Lane 2
7	Lane2_P	True Signal Link Lane 2
8	H_GND	High Speed Ground
9	Lane1_N	Comp Signal Lane 1
10	Lane1_P	True Signal Link Lane 1
11	H_GND	High Speed Ground
12	Lane0_N	Comp Signal Link Lane 0
13	Lane0_P	True Signal Link Lane 0
14	H_GND	High Speed Ground
15	AUX_CH_P	True Signal Auxiliary Ch.
16	AUX_CH_N	Comp Signal Auxiliary Ch.
17	H_GND	High Speed Ground
18	LCD_VCC	LCD logic and driver power
19	LCD_VCC	LCD logic and driver power
20	LCD_VCC	LCD logic and driver power
21	LCD_VCC	LCD logic and driver power
22	LCD_Self TEST	LCD_Self TEST
23	LCD GND	LCD logic and driver ground
24	LCD GND	LCD logic and driver ground
25	LCD GND	LCD logic and driver ground
26	LCD GND	LCD logic and driver ground
27	HPD	HPD signale pin
28	BL_GND	Backlight_ground
29	BL_GND	Backlight_ground
30	BL_GND	Backlight_ground
31	BL_GND	Backlight_ground
32	BL_Enable	Backlight On / Off
33	BL PWM DIM	System PWM signal Input
34	DDS_SDA	DDS I2C SDA
35	NC	NC(Reserve for LCD use)
36	BL_PWR	Backlight power (5V~21V)
37	BL_PWR	Backlight power (5V~21V)
38	BL_PWR	Backlight power (5V~21V)
39	BL_PWR	Backlight power (5V~21V)
40	WP	LCM EEPROM Writing protection pin(Note3)_input voltage 1.8V,serial 1kohm



Note1: Start from right side.

Note2: Input signals shall be low or High-impedance state when VDD is off.
Internal circuit of **eDP inputs** are as following.



6.3 Interface Timing

6.3.1 Timing Characteristics

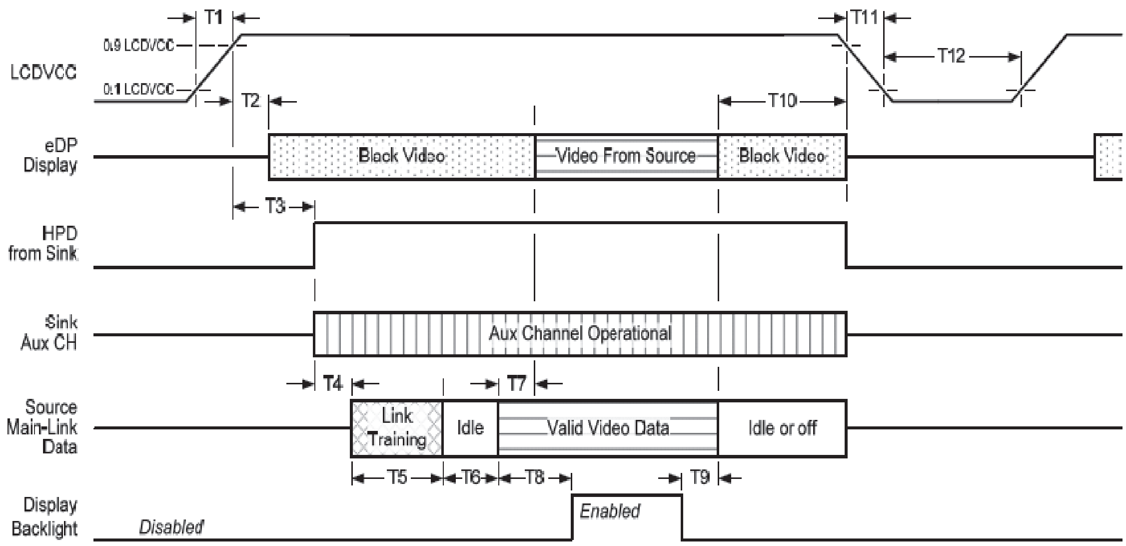
For normal display, interface timings should match the 2560 x 1600 /120Hz manufacturing guide line timing.

Parameter		Symbol	Min.	Typ.	Max.	Unit
Frame Rate		-	60	120	-	Hz
Clock frequency		$1/T_{Clock}$	276.8	553.6	-	MHz
Vertical Section	Period	T_V	1696	1696	1600+A	T_{Line}
	Active	T_{VD}		1600		
	Blanking	T_{VB}	96	96	A	
Horizontal Section	Period	T_H	2720	2720	2560+B	T_{Clock}
	Active	T_{HD}		2560		
	Blanking	T_{HB}	160	160	B	

Note 1 : The above is as optimized setting
Note 2 : The maximum clock frequency = (2560+B)*(1600+A)*120<300MHz

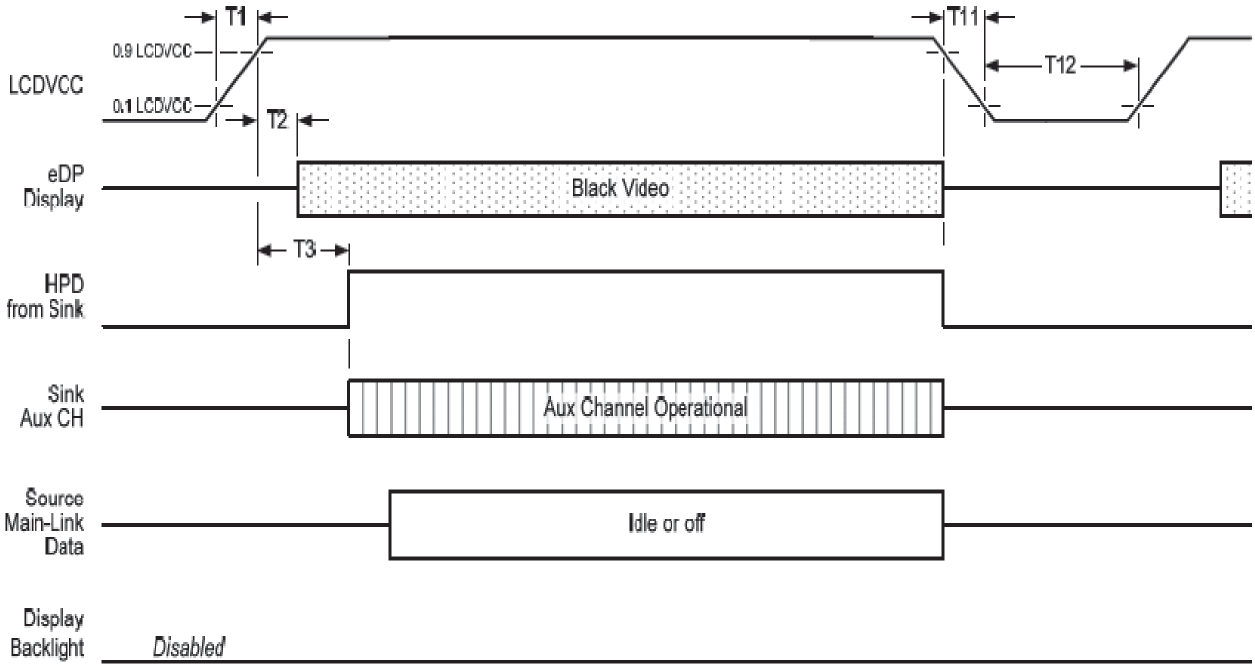
6.4 Power ON/OFF Sequence

Display Port panel power sequence:



Display port interface power up/down sequence, normal system operation

Display Port AUX_CH transaction only:



Display port interface power up/down sequence, AUX_CH transaction only

Display Port panel power sequence timing parameter:

Timing parameter	Description	Reqd. by	Limits			Notes
			Min.	Typ.	Max.	
T1	power rail rise time, 10% to 90%	source	0.5ms		10ms	
T2	delay from LCDVDD to black video generation	sink	0ms		200ms	prevents display noise until valid video data is received from the source
T3	delay from LCDVDD to HPD high	sink	0ms		200ms	sink AUX_CH must be operational upon HPD high.
T4	delay from HPD high to link training initialization	source				allows for source to read link capability and initialize.
T5	link training duration	source				dependant on source link to read training protocol.
T6	link idle	source				Min accounts for required BS-Idle pattern. Max allows for source frame synchronization.
T7	delay from valid video data from source to video on display	sink	0ms		50ms	max allows sink validate video data and timing.
T8	delay from valid video data from source to backlight enable	source				source must assure display video is stable.
T9	delay from backlight disable to end of valid video data	source				source must assure backlight is no longer illuminated.
T10	delay from end of valid video data from source to power off	source	50ms		500ms	
T11	power rail fall time, 905 to 10%	source			10ms	
T12	power off time	source	500ms			

Note1: The sink must include the ability to generate black video autonomously. The sink must automatically enable black video under the following conditions:

-upon LCDVDD power on (with in T2 max)-when the "Novideostream_Flag" (VB-ID Bit 3) is received from the source (at the end of T9).

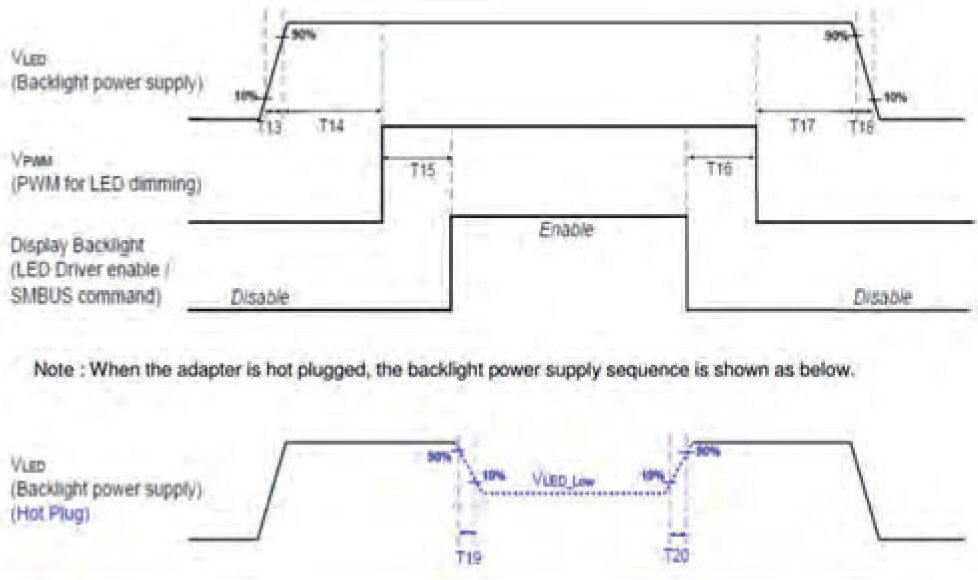
-when no main link data, or invalid video data, is received from the source. Black video must be displayed within 64ms (typ) from the start of either condition. Video data can be deemed invalid based on MSA and timing information, for example.

Note 2: The sink may implement the ability to disable the black video function, as described in Note 1, above, for system development and debugging purpose.

Note 3: The sink must support AUX_CH polling by the source immediately following LCDVDD power on without causing damage to the sink device (the source can re-try if the sink is not ready). The sink must be able to respond to an AUX_CH transaction with the time specified within T3 max.

Note 4:T8>T7

Display Port panel B/L power sequence timing parameter:



	Min (ms)	Max (ms)
T13	0.5	10
T14	10	-
T15	0	-
T16	0	-
T17	10	-
T18	0.5	10
T19	1*	-
T20	1*	-

Seamless change: T19/T20 = $5xT_{PWM}$*

*T_{PWM}= 1/PWM Frequency

7. Panel Reliability Test

7.1 Vibration Test

Test Spec:

- Test method: Non-Operation
- Acceleration: 1.5 G
- Frequency: 10 - 500Hz Random
- Sweep: 30 Minutes each Axis (X, Y, Z)

7.2 Shock Test

Test Spec:

- Test method: Non-Operation
- Acceleration: 220 G , Half sine wave
- Active time: 2 ms
- Pulse: X,Y,Z .one time for each side

7.3 Reliability Test

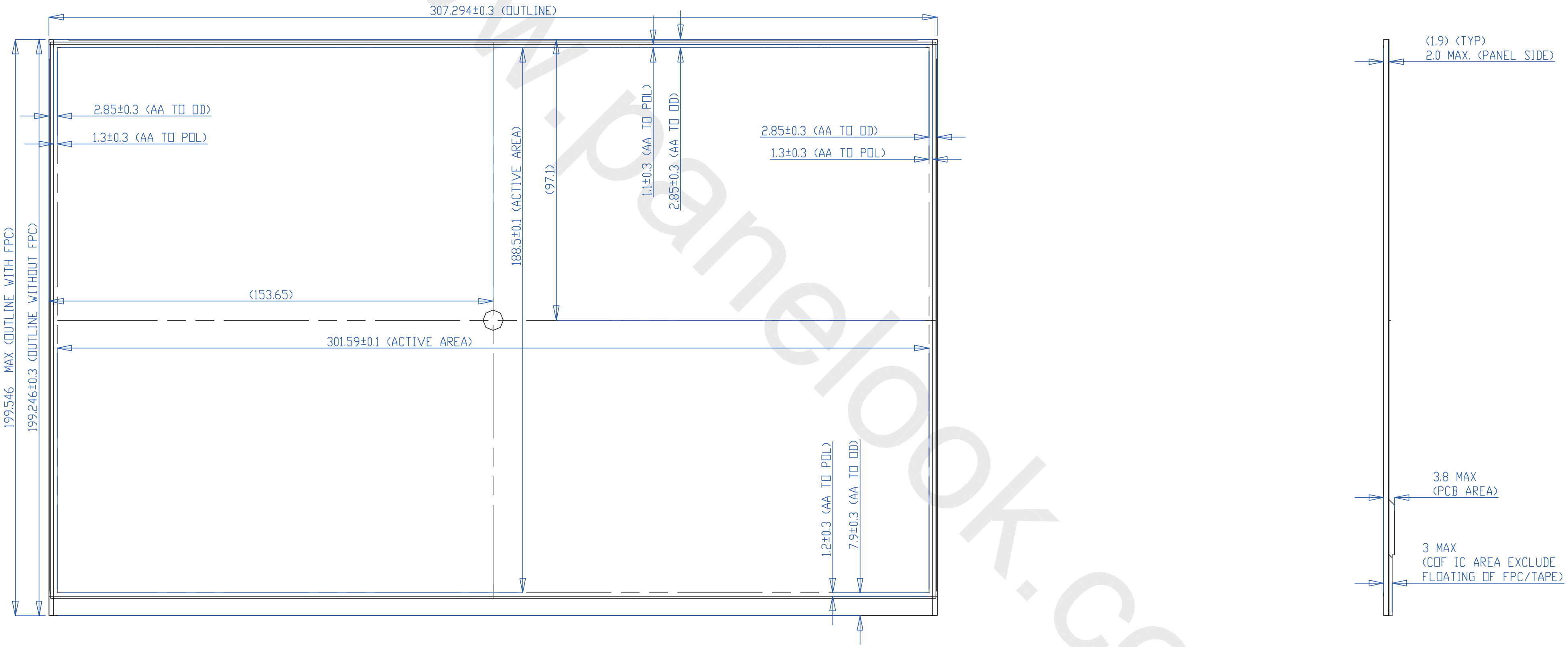
Items	Required Condition	Note
Temperature Humidity Bias	Ta= 40℃, 90%RH, 300h	
High Temperature Operation	Ta= 50℃, Dry, 300h	
Low Temperature Operation	Ta= 0℃, 300h	
High Temperature Storage	Ta= 60℃, 35%RH, 300h	
Low Temperature Storage	Ta= -20℃, 50%RH, 250h	
Thermal Shock Test	Ta=-20℃to 60℃, Duration at 30 min, 100 cycles	
ESD	Contact : ±8 KV Air : ±15 KV	Note 1

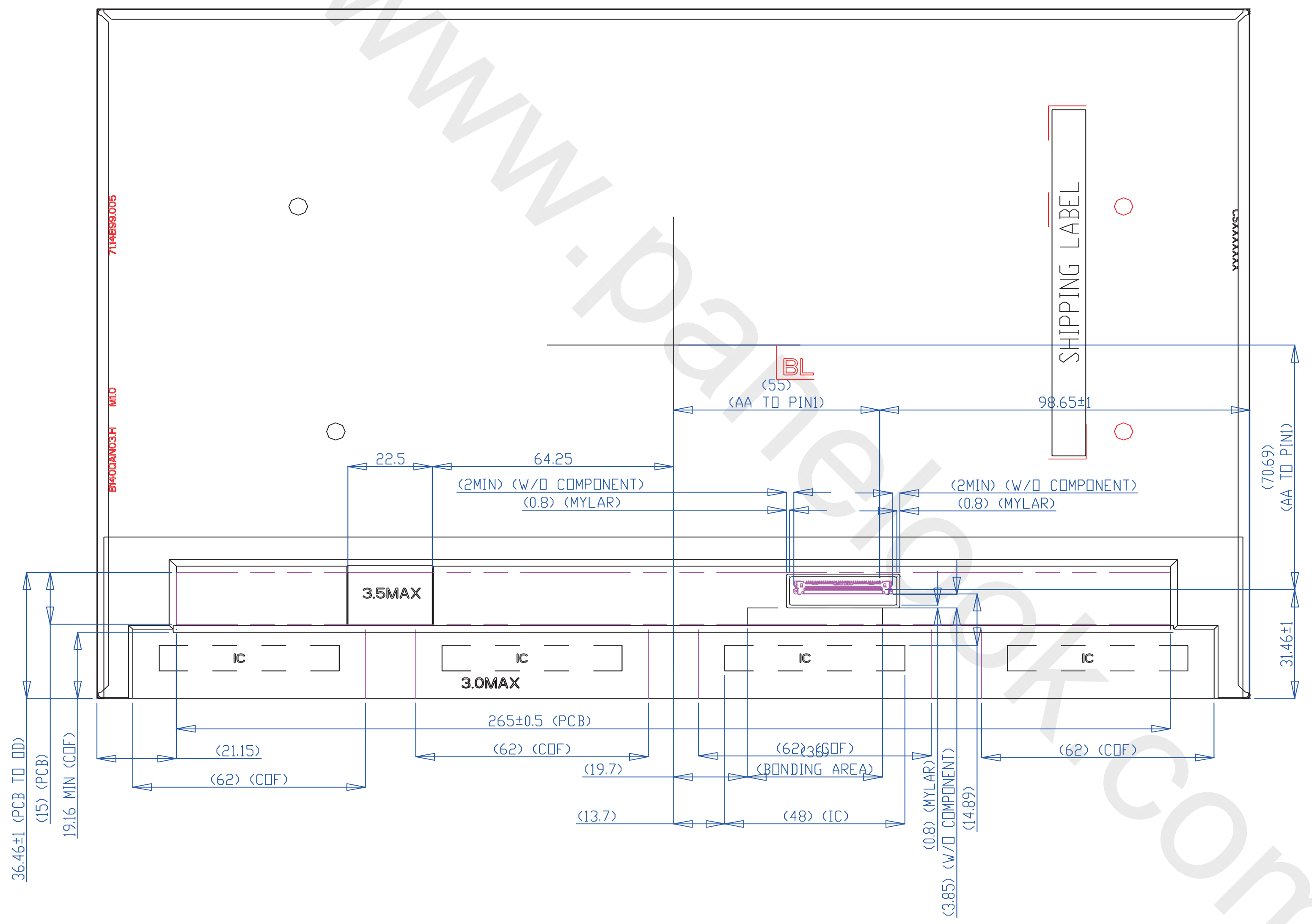
Note1: According to EN 61000-4-2 , ESD class B: Some performance degradation allowed. Self-recoverable. No data lost, No hardware failures.

Remark: MTBF (Excluding the LED): 30,000 hours with a confidence level 90%

8. Mechanical Characteristics

8.1 LCM Outline Dimension



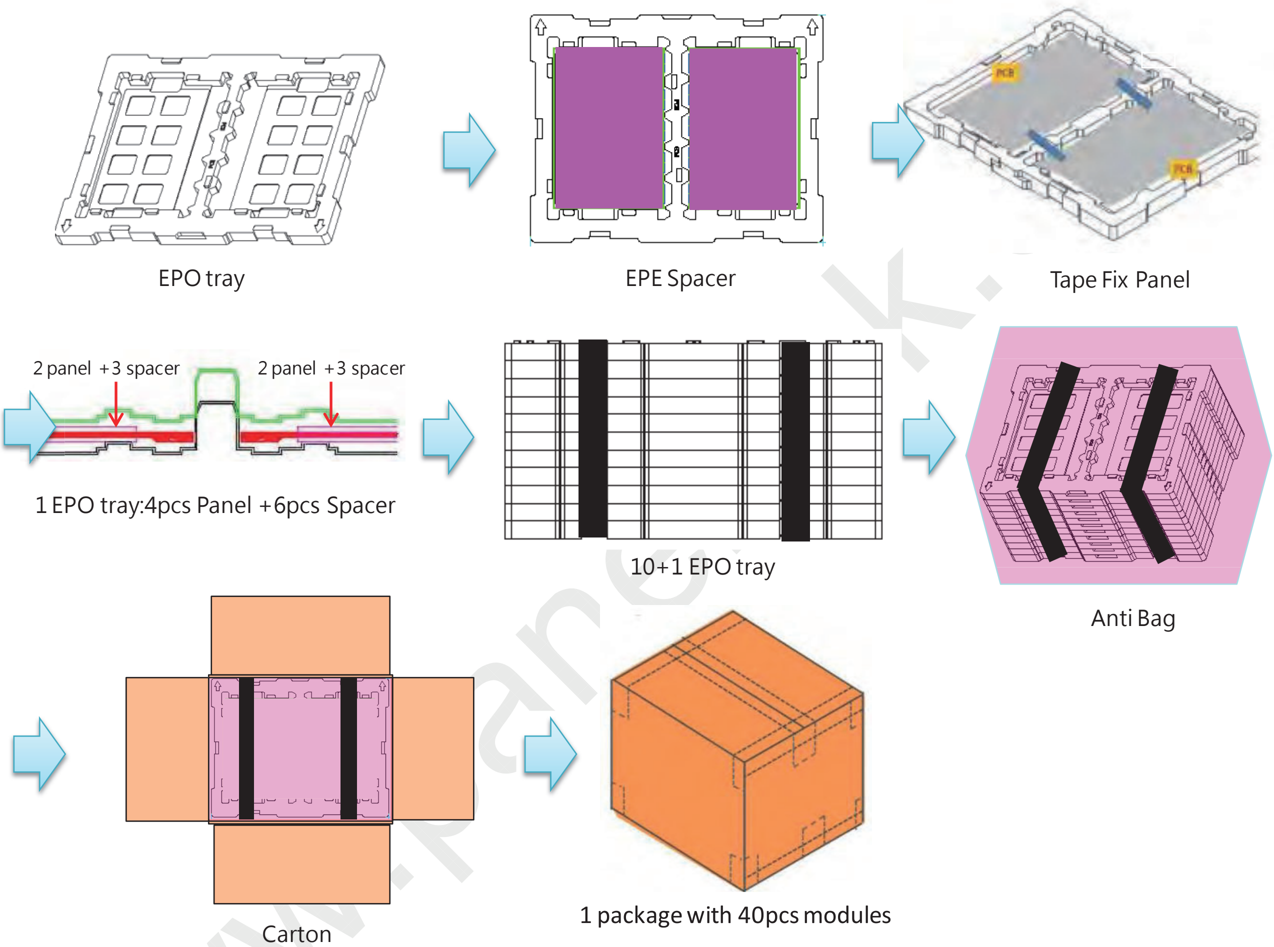


Note: Prevention IC damage, IC positions not allowed any overlap over these areas.

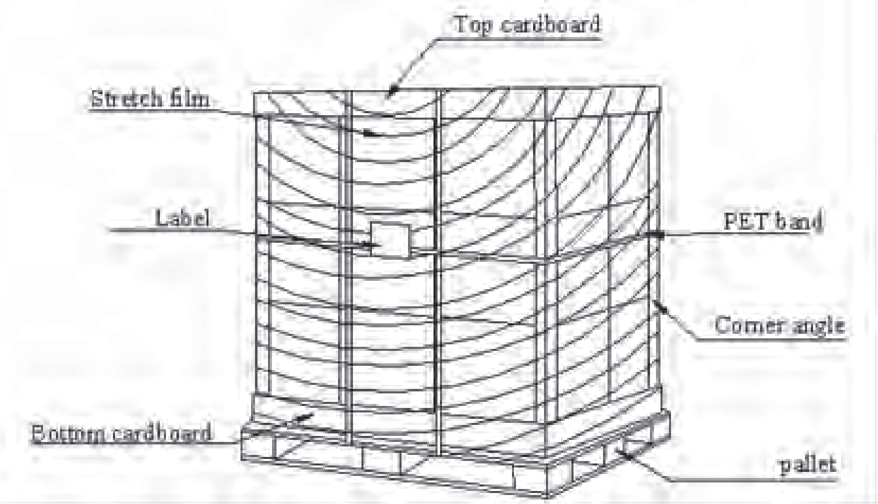
9. Shipping and Package

9.1 Shipping Label Format

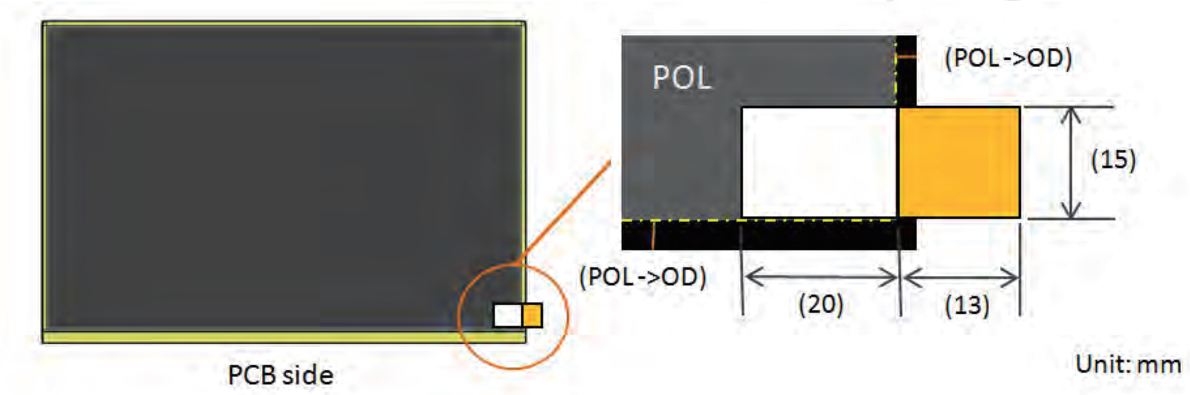
9.2 Carton Package



9.3 Shipping Package of Palletizing Sequence



Pull Bar Position:



10. Appendix:

10.1 EDID Description

	Byte (HEX)	Field Name and Comments	Value (HEX)	Value (Bin)	Value (Dec)
Header	00	Header	00	00000000	0
	01	Header	FF	11111111	255
	02	Header	FF	11111111	255
	03	Header	FF	11111111	255
	04	Header	FF	11111111	255
	05	Header	FF	11111111	255
	06	Header	FF	11111111	255
	07	Header	00	00000000	0
Vendor / Product EDID Version	08	ID Manufacture Name	06	00000110	6
	09	ID Manufacture Name	AF	10101111	175
	0A	ID Product Code	A8	10101000	168
	0B	ID Product Code (Hex. LSB first)	6D	01101101	109
	0C	ID Serial No. - Optional ("00h" If not used, Number Only and LSB First)	00	00000000	0
	0D	ID Serial No. - Optional ("00h" If not used, Number Only and LSB First)	00	00000000	0
	0E	ID Serial No. - Optional ("00h" If not used, Number Only and LSB First)	00	00000000	0
	0F	ID Serial No. - Optional ("00h" If not used, Number Only and LSB First)	00	00000000	0
	10	Week of Manufacture - Optinal	00	00000000	0
	11	Year of Manufacture	20	00100000	32
	12	EDID structure version #	01	00000001	1
	13	EDID revision #	04	00000100	4
Display Parameters	14	Video input Definition = Input is a Digital Video signal Interface , Colo Bit Depth : 8 Bits per Primary Color , Digital Video Interface Standard Supported: DisplayPort is supported	A5	10100101	165
	15	Horizontal Screen Size (Rounded cm)	1E	00011110	30
	16	Vertical Screen Size (Rounded cm)	13	00010011	19
	17	Display Transfer Characteristic (Gamma) = (gamma*100)-100 = Example:(2.2*100)-100=120	78	01111000	120
	18	Feature Support [Display Power Management(DPM) : Standby Mode is not supported, Suspend Mode is not supported, Active Off = Very Low Power is not supported ,Supportted Color Encoding Formats : RGB 4:4:4 ,Other Feature Support Flags : sRGB, Preferred Timing Mode, Display is continuous frequency (Display Range Limits Descriptor is required).]	03	00000011	3
Panel Color Coordinates	19	Red/Green Low Bits (RxRy/GxGy)	E0	11100000	224
	1A	Blue/White Low Bits (BxBy/WxWy)	35	00110101	53
	1B	Red X Rx =	AD	10101101	173
	1C	Red Y Ry =	51	01010001	81
	1D	Green X Gx =	44	01000100	68
	1E	Green Y Gy =	B2	10110010	178
	1F	Blue X Bx =	27	00100111	39
	20	Blue Y By =	0C	00001100	12
	21	White X Wx = 0.313	50	01010000	80
	22	White Y Wy = 0.329	54	01010100	84
Established Timings	23	Established timing 1 (Optional_00h if not used)	00	00000000	0
	24	Established timing 2 (Optional_00h if not used)	00	00000000	0
	25	Manufacturer's timings (Optional_00h if not used)	00	00000000	0

Standard Timing ID	26	Standard timing ID1 (Optional_01h if not used)	01	00000001	1
	27	Standard timing ID1 (Optional_02h if not used)	01	00000001	1
	28	Standard timing ID1 (Optional_03h if not used)	01	00000001	1
	29	Standard timing ID1 (Optional_04h if not used)	01	00000001	1
	2A	Standard timing ID1 (Optional_05h if not used)	01	00000001	1
	2B	Standard timing ID1 (Optional_06h if not used)	01	00000001	1
	2C	Standard timing ID1 (Optional_07h if not used)	01	00000001	1
	2D	Standard timing ID1 (Optional_08h if not used)	01	00000001	1
	2E	Standard timing ID1 (Optional_09h if not used)	01	00000001	1
	2F	Standard timing ID1 (Optional_10h if not used)	01	00000001	1
	30	Standard timing ID1 (Optional_11h if not used)	01	00000001	1
	31	Standard timing ID1 (Optional_12h if not used)	01	00000001	1
	32	Standard timing ID1 (Optional_13h if not used)	01	00000001	1
	33	Standard timing ID1 (Optional_14h if not used)	01	00000001	1
	34	Standard timing ID1 (Optional_13h if not used)	01	00000001	1
	35	Standard timing ID1 (Optional_14h if not used)	01	00000001	1
Timing Description #1 (preferred fresh rate)	36	Pixel Clock/10,000 (LSB)	3E	00111110	62
	37	Pixel Clock/10,000 (MSB)	D8	11011000	216
	38	Horizontal Active (HA) (lower 8 bits)	00	00000000	0
	39	Horizontal Blanking (HB) (lower 8 bits)	A0	10100000	160
	3A	Horizontal Active (HA) / Horizontal Blanking (HB) (upper 4:4bits)	A0	10100000	160
	3B	Vertical Avtive (VA)	40	01000000	64
	3C	Vertical Blanking (VB) (DE Blanking typ.for DE only panels)	60	01100000	96
	3D	Vertical Active (VA) / Vertical Blanking (VB) (upper 4:4bits)	60	01100000	96
	3E	Horizontal Front Porch in pixels (HF) (lower 8 bits)	30	00110000	48
	3F	Horizontal Sync Pulse Width in pixels (HS) (lower 8 bits)	20	00100000	32
	40	Vertical Front Porch in lines (VF) : Vertical Sync Pluse Width in lines (VS) (lower 4 bits)	35	00110101	53
	41	Horizontal Front Porch/ Sync Pulse Width/ Vertical Front Porch/ Sync Pulse Width (upper 2bits)	00	00000000	0
	42	Horizontal Vedio Image Size (mm) (lower 8 bits)	2D	00101101	45
	43	Vertical Vedio Image Size (mm) (lower 8 bits)	BC	10111100	188
	44	Horizontal Image Size / Vertical Image Size (upper 4 bits)	10	00010000	16
	45	Horizontal Border = 0 (Zero for Notebook LCD)	00	00000000	0
	46	Vertical Border = 0 (Zero for Notebook LCD)	00	00000000	0
	47	Non-Interlace, Normal display, no stereo, Digital Separate [Vsync_NEG, Hsync_POS (outside of V-sync)]	18	00011000	24
Timing Description #2	48	Pixel Clock/10,000 (LSB)	20	00100000	32
	49	Pixel Clock/10,000 (MSB)	6C	01101100	108
	4A	Horizontal Active (HA) (lower 8 bits)	00	00000000	0
	4B	Horizontal Blanking (HB) (lower 8 bits)	A0	10100000	160
	4C	Horizontal Active (HA) / Horizontal Blanking (HB) (upper 4:4bits)	A0	10100000	160
	4D	Vertical Avtive (VA)	40	01000000	64
	4E	Vertical Blanking (VB) (DE Blanking typ.for DE only panels)	60	01100000	96
	4F	Vertical Active (VA) / Vertical Blanking (VB) (upper 4:4bits)	60	01100000	96
	50	Horizontal Front Porch in pixels (HF) (lower 8 bits)	30	00110000	48
	51	Horizontal Sync Pulse Width in pixels (HS) (lower 8 bits)	20	00100000	32

	52	Vertical Front Porch in lines (VF) : Vertical Sync Pluse Width in lines (VS) (lower 4 bits)	35	00110101	53
	53	Horizontal Front Porch/ Sync Pulse Width/ Vertical Front Porch/ Sync Pulse Width (upper 2bits)	00	00000000	0
	54	Horizontal Vedio Image Size (mm) (lower 8 bits)	2D	00101101	45
	55	Vertical Vedio Image Size (mm) (lower 8 bits)	BC	10111100	188
	56	Horizontal Image Size / Vertical Image Size (upper 4 bits)	10	00010000	16
	57	Horizontal Border = 0 (Zero for Notebook LCD)	00	00000000	0
	58	Vertical Border = 0 (Zero for Notebook LCD)	00	00000000	0
	59	Non-Interlace, Normal display, no stereo, Digital Separate [Vsync_NEG, Hsync_POS (outside of V-sync)]	18	00011000	24
Timing Description #3 (Brightness Table)	5A		00	00000000	0
	5B		00	00000000	0
	5C		00	00000000	0
	5D		00	00000000	0
	5E		00	00000000	0
	5F		00	00000000	0
	60		00	00000000	0
	61		00	00000000	0
	62		00	00000000	0
	63		00	00000000	0
	64		00	00000000	0
	65		00	00000000	0
	66		00	00000000	0
	67		00	00000000	0
	68		00	00000000	0
	69		00	00000000	0
	6A		00	00000000	0
	6B		00	00000000	0
Timing Description #4 (Brightness Table)	6C	Detailed Timing Description #4	00	00000000	0
	6D	Flag	00	00000000	0
	6E	Reserved	00	00000000	0
	6F	For Brightness Table and Power Consumption	02	00000010	2
	70	Flag	00	00000000	0
	71	PWM % [7:0] @ Step 0 or FW version for OLED	0C	00001100	12
	72	PWM % [7:0] @ Step 5 or FW version for OLED	24	00100100	36
	73	PWM % [7:0] @ Step 10 or FW version for OLED	FF	11111111	255
	74	Nits [7:0] @ Step 0	13	00010011	19
	75	Nits [7:0] @ Step 5	3C	00111100	60
	76	Nits [7:0] @ Step 10	FA	11111010	250
	77	Panel Electronics Power @ 32 x 32 Chess Pattern =	1E	00011110	30
	78	Backlight Power @ 60 nits =	10	00010000	16
	79	Backlight Power @ Step 10 =	46	01000110	70
	7A	Nits [7:0] @ 100% PWM Duty =	FA	11111010	250

	7B	Flags	20	00100000	32
	7C	Flags	20	00100000	32
	7D	Flags	20	00100000	32
ChkSum	7E	Extension flag	01	00000001	1
	7F	Check Sum	3F	00111111	63
				SUM	6912

Extend EDID

B140QAN03.H Extend EDID Code

Hex Address	Field Name	Value Hex	Value (dec)	Value (bin)
80	**EDID extension block tag**	70	112	01110000
81	DisplayID Version/Revsion = 2.0	20	32	00100000
82	Section Size (byte) =	79	121	01111001
83	Display Product Primary Use Case	02	2	00000010
84	Extension count	00	0	00000000
85	DID2.0 Data block tag[22h] =Type VII Timing – Detailed Timing Data Block	22	34	00100010
86	Block Revision and Other Data	00	0	00000000
87	Number of Payload Bytes in Block	14	20	00010100
88	Timing 1 Pixel Clock (in kHz) Rate range is defined as 0.001 through 16,777.216MP/s.	6B	107	01101011
89		72	114	01110010
8A		08	8	00001000
8B	Timing Options	84	132	10000100
8C	Horizontal Active Image Pixels Number of pixels ranges from 1 through 65,536.	FF	255	11111111
8D		09	9	00001001
8E	Horizontal Blank Pixels Number of pixels ranges from 1 through 65,536.	9F	159	10011111
8F		00	0	00000000
90	Horizontal Offset (Front Porch) Number of pixels ranges from 1 through 32,768.	2F	47	00101111
91		80	128	10000000
92	Horizontal Sync Width Number of pixels ranges from 1 through 65,536.	1F	31	00011111
93		00	0	00000000
94	Vertical Active Image Lines Number of lines ranges from 1 through 65,536.	3F	63	00111111
95		06	6	00000110
96	Vertical Blank Lines Number of lines ranges from 1 through 65,536.	5F	95	01011111
97		00	0	00000000
98	Vertical Sync Offset (Front Porch) Number of lines ranges from 1 through 32,768.	02	2	00000010
99		00	0	00000000
9A	Vertical Sync Width Number of lines ranges from 1 through 65,536.	04	4	00000100
9B		00	0	00000000

9C	**DID2.0 Data block tag[25h] = Dynamic Video Timing Range Limits**	25	37	00100101
9D	Block revision = Revision 1	01	1	00000001
9E	Number of Payload Bytes in block=	09	9	00001001
9F	Minimum Pixel Clock (Low bit, Range = 0.001Mhz (000000h) ~ 16,777.216Mhz (FFFFFFh))	6B	107	01101011
A0	Minium Pixel Clock (Middle bit)	72	114	01110010
A1	Minium Pixel Clock (High bit)	08	8	00001000
A2	Maximum Pixel Clock (Low bit, Range = 0.001Mhz (000000h) ~ 16,777.216Mhz (FFFFFFh))	6B	107	01101011
A3	Maximum Pixel Clock (Middle bit)	72	114	01110010
A4	Maximum Pixel Clock (High bit)	08	8	00001000
A5	Min. Vertical Rate : ?? Hz (Range : 0Hz (00h) ~ 255Hz (FFh)	30	48	00110000
A6	Max. Vertical Rate : ?? Hz (Range : 0Hz (000h) ~ 1023Hz (3FFh))	78	120	01111000
A7	Seamless Dynamic Video Timing Support : Seamless Dynamic Video Timing change shall be supported with a fixed horizontal pixel rate and dynamic vertical blanking.	80	128	10000000
A8	**DID2.0 Data block tag[81h] = CTA DisplayID**	81	129	10000001
A9	Block revision	00	0	00000000
AA	Number of Payload Bytes in block=	10	16	00010000
AB	CTA Block Tag Code and Block Length = Vendor Specific Data Block(03h) +Length of following data block (in bytes)	72	114	01110010
AC	AMD IEEE OUI value (0x00001A)	1A	26	00011010
AD	(Hex. LSB first)	00	0	00000000
AE	(Hex. LSB first)	00	0	00000000
AF	AMD VSDB Version 3	03	3	00000011
B0	Bit 0 = 1; Seamless Variable Frame Rate Switching Supported Bit 1 = 0; Seamless Native Color Space & Transfer Switching Curve Non-Supported Bit 2 = 0 Bit 3 = 0; Seamless Local Dimming Disable Control Non-Supported Bit 5:7 = 0	01	1	00000001
B1	Min Refresh Rate	30	48	00110000
B2	Max Refresh Rate	78	120	01111000
B3	0x00 or MCCS code to query current state of FreeSync.	00	0	00000000
B4	Supported WCG and HDR features Bit 2 = Gamma 2.2 Supported (mandatory) Bit 5 = PQ EOTF Supported	00	0	00000000
B5	Max Luminance 1 (for HDR) =	6A	106	01101010
B6	Min Luminance 1 (for HDR) =	49	73	01001001
B7	Max Luminance 2 (for HDR) =	6A	106	01101010
B8	Min Luminance 2 (for HDR) =	49	73	01001001
B9	Freesync Maximum Refresh Rate (LSB) : (Range : 0Hz (000h) ~ 1023Hz (3FFh), for VSDB v3)	78	120	01111000
BA	Freesync Maximum Refresh Rate (MSB)	00	0	00000000
BB~FD		00	0	00000000
FE	Section Checksum (Mandatory)	DF	223	11011111
FF	EDID Extension Block Checksum (Mandatory)	90	144	10010000

10.2 Notes

DPCD Ver.	sDRRS	DCR	DMRRS	PSR	LRR	LRR Frame rate	MBO	VESA DSC
1.4	ON	NA	NA	PSR2 on	LRR1.0	60Hz	NA	NA

MSO	G-Sync	Free-Sync	Adaptive Sync	HDR Tier	HDR Ver.	System Input
NA	NA	ON MRL 48-120hz	NA	NA	NA	PWM

ASSR	Auto ASSR (Tcon support)	VSC/SDP (HDR / eDP1.4up)	Dream color + Flash lock	WP (Flash lock)
ON	NA	NA	ON	ON